

Forecast Settings

Upload PJM dataset

PJMW_MW_Hourly.xlsx
2.5MB

+

Forecast horizon (days)

30

-

+

Final notebook model: XGBoost

Forecast horizon: 30 days / 720 hours



PJM Hourly Energy Consumption Forecast

P-702 / P679 Forecasting Project

This application forecasts hourly PJM electricity demand using the **XGBoost model selected in the project notebook**.

1. Dataset Overview

Raw observations	Start	End	Average Load
143,206	2002-04-01	2018-08-03	5,602.38 ...

The notebook dataset contains the columns Datetime and PJMW_MW. Feature engineering follows the final notebook workflow.

2. Feature Engineering

Modeling observations	Forecast features	US holiday dates	Frequency
143,038	16	189	Hourly

View feature list

▼

[
 0 : "Hour"
 1 : "Day"
 2 : "DayOfWeek"
 3 : "Month"
 4 : "Year"
 5 : "IsWeekend"
 6 : "IsHoliday"
 7 : "lag_1"
 8 : "lag_2"
 9 : "lag_3"
 10 : "lag_24"
 11 : "lag_48"
 12 : "lag_168"
 13 : "rolling_mean_24"
 14 : "rolling_mean_168"
 15 : "rolling_std_24"
]

3. Train / Test Split

Training: 2002-04-08 to 2017-08-02

Testing: 2017-08-03 to 2018-08-03

Training observations: 134,277

Testing observations: 8,761

4. Final Forecasting Model

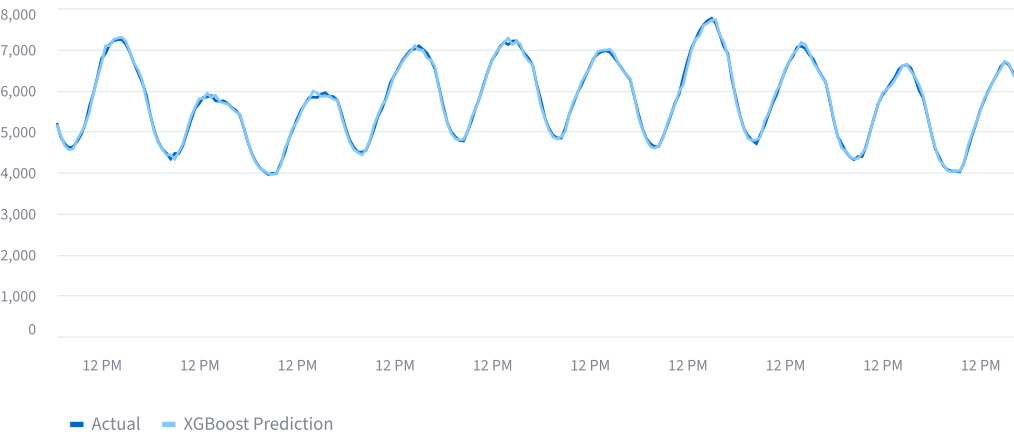
XGBoost Regressor — selected as the final model in the notebook because it provided the lowest overall prediction errors.

```
XGBRegressor(  
    n_estimators=500,  
    learning_rate=0.05,  
    max_depth=7,  
    min_child_weight=1,  
    subsample=0.8,  
    colsample_bytree=0.8,  
    objective="reg:squarederror",  
    random_state=42  
)
```

5. Model Performance on Test Set

MAE	RMSE	MAPE	R ²
52.09 MW	70.28 MW	0.91%	0.9950

6. Actual vs XGBoost Prediction



7. XGBoost Feature Importance

Feature	Importance
lag_1	
lag_2	
lag_24	
lag_168	
lag_3	
Hour	
IsWeekend	
DayOfWeek	
rolling_mean_24	
rolling_std_24	

8. 30-Day Hourly Forecast

Generate Forecast

Forecast generated successfully for 720 hourly observations.

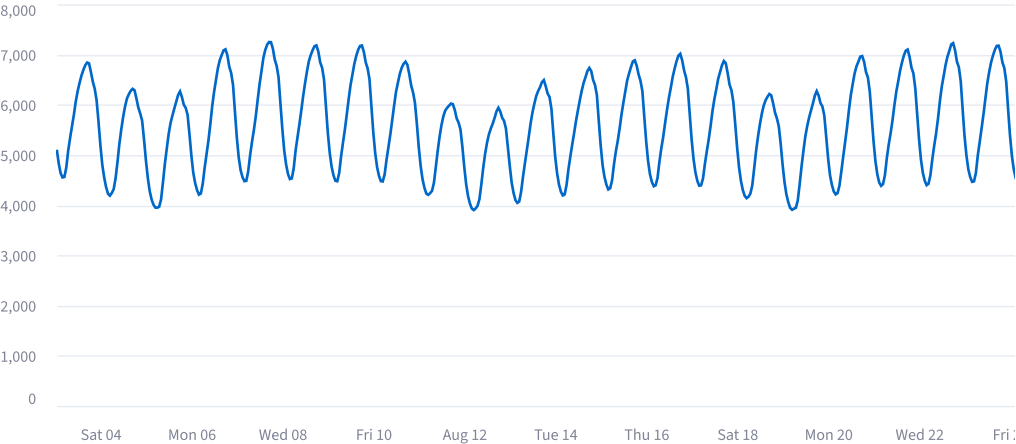
Average Forecast Peak Forecast Minimum Forecast

5,552.40 MW

7,250.28 MW

3,881.80 MW

9. Forecast Visualization



10. Hourly Forecast Table

Datetime	Predicted_MW
2018-08-03 01:00:00	
2018-08-03 02:00:00	
2018-08-03 03:00:00	
2018-08-03 04:00:00	
2018-08-03 05:00:00	
2018-08-03 06:00:00	
2018-08-03 07:00:00	
2018-08-03 08:00:00	
2018-08-03 09:00:00	
2018-08-03 10:00:00	

11. Daily Forecast Summary

Forecast_Day	Start_Datetime	End_Datetime	Average_Forecast_MW	Minimum_Forecast_MW
1	2018-08-03 01:00:00	2018-08-04 00:00:00	5796.32	
2	2018-08-04 01:00:00	2018-08-05 00:00:00	5314.51	
3	2018-08-05 01:00:00	2018-08-06 00:00:00	5120.95	
4	2018-08-06 01:00:00	2018-08-07 00:00:00	5767.04	
5	2018-08-07 01:00:00	2018-08-08 00:00:00	5965.58	
6	2018-08-08 01:00:00	2018-08-09 00:00:00	5973.83	
7	2018-08-09 01:00:00	2018-08-10 00:00:00	5934.75	
8	2018-08-10 01:00:00	2018-08-11 00:00:00	5736.85	
9	2018-08-11 01:00:00	2018-08-12 00:00:00	5175.07	
10	2018-08-12 01:00:00	2018-08-13 00:00:00	4948.8	

Download Hourly Forecast CSV

Download Daily Summary CSV

